

Sasikumar Karikalan

Full Stack Developer & Quality Assurance Engineer



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PROFESSIONAL SUMMARY

Dynamic Full Stack Developer and QA Engineer with 15+ years of experience, including **11.6 years at Wipro Technologies with 6 years of onsite experience in Boston, USA**, followed by **3+ years as an Independent Tech Developer through my registered company**, delivering solutions for U.S. clients. Proven expertise in JavaScript, TypeScript, React, Node.js, and Express.js, with a strong foundation in Agile methodologies, test automation (Selenium, Postman), CI/CD pipelines, and AWS. Recognized for leading mission-critical projects that scaled products by 30X, delivering high-quality solutions across eCommerce, education, and enterprise platforms while driving innovation, performance, and client success.

SKILLS

Programming Languages : JavaScript, TypeScript, Node.js, HTML, CSS, Java

Frameworks/Libraries : React, Express.js, Nextjs, TailwindCSS, Bootstrap

Database Management : MySQL, MongoDB

Testing & QA Automation Tools : JIRA, Selenium WebDriver, JUnit, TestNG, Maven

API Testing Tools : Postman, RestAssured

Build & CI/CD Tools : Jenkins, CircleCI, GitLab CI/CD

Cloud Platforms : AWS (S3, EC2, Lambda, RDS)

Version Control : Git, GitHub

IDEs & Other Tools : Visual Studio Code (VSCode), Eclipse

EXPERIENCE

FULL STACK DEVELOPER, Chennai, TamilNadu

Freelancer, May 2024–Present

I am currently contributing to building and maintaining web applications using a modern JavaScript/TypeScript stack, working on both client-side and server-side development to ensure seamless integration and high performance across the application.

- **Front-End:** Developing dynamic and responsive UI components using **React** and **TypeScript**, following best practices for clean, modular code. I implement state management using React hooks and context API and optimize performance for a smoother user experience.

- **Back-End:** Designing and building **RESTful APIs** using **Node.js** and **Express.js**, ensuring secure, scalable, and efficient server-side operations. I manage authentication, data validation, and business logic implementation.
- **Database Integration:** Integrating the back-end with databases (**MongoDB/MySQL**), handling CRUD operations and ensuring proper data flow between the front-end and back-end.
- **Testing & Debugging:** Utilizing Postman for unit testing and API testing, ensuring robust functionality across the stack. I collaborate with the QA team to resolve bugs and improve application reliability.
- **Version Control & CI/CD:** Working with **Git** and **GitHub** for version control and managing CI/CD pipelines using Jenkins to automate testing and deployment.

I collaborate closely with designers, product managers, and fellow developers, following Agile methodologies to deliver features in an iterative and timely manner.

VR 3D APPLICATION TESTER AND DEVELOPER, Chennai, TamilNadu

Freelancer, March 2023–April 2024

Worked with a U.S.- based client on developing and testing **Virtual Reality(VR) 3D music applications** using **Unity**. Focused on testing core functionalities, addressing bugs, and ensuring smooth user experiences in an immersive environment. Contributed to the development of interactive features, troubleshooting performance issues, and improving app responsiveness. Assisted in the design and implementation of user interfaces, scene transitions, and audio integration. Collaborated closely with the client to meet project milestones, provide technical solutions, and enhance overall application quality. Successfully delivered a well-optimized VR product tailored to the client's requirements.

LEAD QUALITY ASSURANCE ENGINEER, Boston, MA

Wipro Technologies, November 2015–November 2021

- **Company :** Wipro Technologies.
- **Client :** McGraw Hill Education, Boston, Massachusetts.
- **Project:** Access Manager - Rostering application for Millions of students and teachers in USA.
- **Role :** Lead Quality Assurance Engineer.
- **Responsibilities :**

- 1) Solely responsible for leading the quality framework of a mission-critical application from inception to maturity.
- 2) Leveraged new tools in **JIRA** to build frameworks that support scalability, utilizing API collections in **Postman** for integrations testing shared across various teams.
- 3) Led a team in transitioning to a microservices **CI/CD** architecture, enabling the team to break release dependencies and collaborate more closely with customers to address issues in near real-time.
- 4) Advocated for and trained a team of quality engineers in functional, automation, and performance testing methodologies.
- 5) Collaborated across teams and programs to lead highly complex integration initiatives for critical access services.
- 6) Worked closely with customers to understand their pain points and developed tools to eliminate bottlenecks during key delivery periods.
- 7) Managed over 50 enhancement releases each year after each Back to School period based on customer requests.
- 8) Led the team in integration testing with a robust test approach and strategy that improved product quality across all teams.
- 9) Oversaw more than 300 releases in those 6 years, coordinating efforts between onshore and offshore teams.
- 10) Conducted **API, ETL, UI, and Microservices** integration testing, implementing a hybrid automation framework model using **Selenium**.

Achievements

- Successfully delivered the application to production in 6 months, scaling it 30X in the first year, increasing user support from 1,800 to 18 million over three years.
- The Access Manager project was recognized by the board of directors as the highest-performing team in the department, delivering on committed scope 98% of the time while maintaining the lowest production bug count across all production applications.
- Received FIVE awards for selfless teamwork and leadership within the team.

QUALITY ASSURANCE ENGINEER, Chennai, TamilNadu

Wipro Technologies, November 2010–October 2015

- **Company** : Wipro Technologies.
- **Clients** : McGraw Hill, NIKE, KOHLS - USA.
- **Role** : Quality Assurance Engineer.
- **Responsibilities** :

- 1) Conducted thorough functional testing on web applications to ensure seamless user experience and smooth functionality across the eCommerce platforms like NIKE and KOHLS.
- 2) Validated end-to-end workflows, including product search, cart management, checkout, and payment gateway integrations.
- 3) Performed testing across multiple browsers (Chrome, FireFox, Safari, IE) and devices (desktop, tablet, mobile) to ensure responsive design and compatibility on various platforms.
- 4) Evaluated the user interface for compliance with design specifications, ensuring the website maintained a consistent, user-friendly experience across all pages.
- 5) Developed automated test scripts using tools like **Selenium** to speed up regression testing and reduce manual effort, ensuring consistency in repetitive tasks.
- 6) Automated key scenarios like user login, order processing, and inventory management to improve testing efficiency.
- 7) Conducted API testing using **Postman** and other tools to validate interactions between frontend and backend systems, ensuring that data flows correctly between the server and client.
- 8) Tested payment gateways, order fulfillment systems, and third-party service integrations.
- 9) Ensured optimal website performance by performing load testing, stress testing, and analyzing site speed during peak traffic periods (sales, special promotions)
- 10) Used tools like **JMeter** to assess system behavior under heavy user loads and reported bottlenecks.
- 11) Used **JIRA** to log defects, track resolutions, and manage test cycles, collaborating with developers to resolve issues and ensure high-quality releases.
- 12) Prioritized and categorized bugs based on severity and impact on the eCommerce platform.
- 13) Designed, documented, and executed detailed test cases for various features of the eCommerce website, covering functional, integration, and regression testing.
- 14) Created test plans aligned with client requirements and business goals.
- 15) Worked closely with developers, project managers, and product teams to ensure test coverage and provide feedback throughout the development cycle.
- 16) Participated in **Agile/Scrum meetings** to plan sprints and ensure timely delivery of features.
- 17) Ensured that all aspects of the eCommerce platform, from product browsing to order completion, worked seamlessly together by conducting thorough end-to-end testing.
- 18) Validated integration with third-party services like payment gateways, shipment tracking, and customer support systems.
- 19) Also used **HP quality Center** to manage test cases, from creation and execution to defect logging, ensuring full traceability of test processes.
- 20) Provided regular updates on test progress, issues, and risks to senior team members and stakeholders, ensuring transparency and effective communication within the team.

EDUCATION

BE IN ELECTRONICS AND COMMUNICATION ENGINEERING, Chennai

Anna University, June 2010

LANGUAGES

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| • Tamil
Native | • English
Fluent |
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